

MSTAR PRACTICE WORKSHEET**Lesson 5-3: Determine the Area of Trapezoids**

Grade 6 Mathematics · Standard 6.GR.1

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: B**

What is the area of a trapezoid with bases 6 cm and 10 cm, and height 4 cm?

- ☐ A 24 sq cm
- ☒ B 32 sq cm
- ☐ C 40 sq cm
- ☐ D 64 sq cm

$$A = \frac{1}{2} \times (b_1 + b_2) \times h = \frac{1}{2} \times (6 + 10) \times 4 = \frac{1}{2} \times 16 \times 4 = 32 \text{ sq cm.}$$

Question 2 **SELECTED RESPONSE — CORRECT: B**

A trapezoid has an area of 45 sq ft, bases of 7 ft and 11 ft. What is the height?

- ☐ A 3 ft
- ☒ B 5 ft
- ☐ C 9 ft
- ☐ D 10 ft

$$A = \frac{1}{2} \times (b_1 + b_2) \times h \rightarrow 45 = \frac{1}{2} \times 18 \times h \rightarrow 45 = 9h \rightarrow h = 5 \text{ ft.}$$

Question 3 SELECTED RESPONSE — CORRECT: C

What is the first step when finding the area of a trapezoid?

- ☐ A Multiply the two bases
- ☐ B Subtract the shorter base from the longer base
- ☒ C Add the two bases together
- ☐ D Divide the height by 2

The formula is $A = \frac{1}{2} \times (b_1 + b_2) \times h$. The first step is to add the two parallel bases.

Question 4 SELECTED RESPONSE — CORRECT: C

A trapezoid has bases of 5 in and 9 in, and a height of 6 in. What is its area?

- ☐ A 27 sq in
- ☐ B 30 sq in
- ☒ C 42 sq in
- ☐ D 84 sq in

$A = \frac{1}{2} \times (5 + 9) \times 6 = \frac{1}{2} \times 14 \times 6 = \frac{1}{2} \times 84 = 42$ sq in.

Question 5 SELECTED RESPONSE — CORRECT: B

Blueprint 1: A trapezoidal lobby window has bases of 6 ft and 10 ft and a height of 4 ft. What is its area for the glass order?

- ☐ A 16 sq ft
- ☒ B 32 sq ft
- ☐ C 40 sq ft
- ☐ D 60 sq ft

Use $A = \frac{1}{2} \times (b_1 + b_2) \times h = \frac{1}{2} \times (6 + 10) \times 4 = \frac{1}{2} \times 16 \times 4 = 32$ sq ft.

Question 6

SELECTED RESPONSE — CORRECT: B

Blueprint 2: A trapezoidal skylight has bases of 9 ft and 7 ft and a height of 6 ft. What is its area?

- ☐ A 44 sq ft
- ☒ B 48 sq ft
- ☐ C 63 sq ft
- ☐ D 96 sq ft

$$A = \frac{1}{2} \times (9 + 7) \times 6 = \frac{1}{2} \times 16 \times 6 = \frac{1}{2} \times 96 = 48 \text{ sq ft.}$$

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7

TWO-PART QUESTION

Part A — correct answer: C

A trapezoidal window has parallel edges of 4 feet and 8 feet and a height of 5 feet. What is its area?

- ☐ A 17 square feet
- ☐ B 20 square feet
- ☒ C 30 square feet
- ☐ D 60 square feet

$$\text{Area} = \frac{1}{2} \times (b_1 + b_2) \times h = \frac{1}{2} \times (4 + 8) \times 5 = \frac{1}{2} \times 12 \times 5 = 30 \text{ square feet.}$$

- **A:** That adds all three measurements. The rule multiplies the average of the parallel edges by the height.
- **B:** That is 4 times 5, using only one of the two parallel edges. Average them first.
- **D:** The halving step was skipped. 4 plus 8, times 5, is twice the area.

Part B — correct answer: A

What does the quantity $(b_1 + b_2) \div 2$ represent in the trapezoid area formula?

- ☒ A The average length of the two parallel bases.
- ☐ B The perimeter of the trapezoid.
- ☐ C The length of the slanted side.
- ☐ D The height of the trapezoid.

Averaging the two bases turns the trapezoid into a rectangle of the same area with that average width, which is then multiplied by the height.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D, E

Select ALL statements that are true about trapezoids and their area:

- ☒ A A trapezoid has exactly one pair of parallel sides called bases.
- ☒ B The height is measured perpendicular to both parallel bases.
- ☐ C The two bases must be equal in length.
- ☒ D A trapezoid can be split into two triangles to find its area.
- ☒ E Area = $\frac{1}{2} \times (b_1 + b_2) \times h$.

A, B, D, and E are true. C is false — if both bases were equal the figure would be a parallelogram.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A trapezoid and a parallelogram both have a height of 8 ft. The trapezoid has bases of 5 ft and 11 ft. The parallelogram has a base of 7 ft. Which shape has the greater area? Explain your reasoning using the formulas.

Model answer: The trapezoid's area is $\frac{1}{2} \times (5 + 11) \times 8 = \frac{1}{2} \times 16 \times 8 = 64$ sq ft and the parallelogram's is $7 \times 8 = 56$ sq ft, so the trapezoid is greater by 8 sq ft. Since both shapes share the height of 8 ft, the comparison really comes down to the trapezoid's average base, $\frac{1}{2} \times (5 + 11) = 8$ ft, against the parallelogram's base of 7 ft.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.